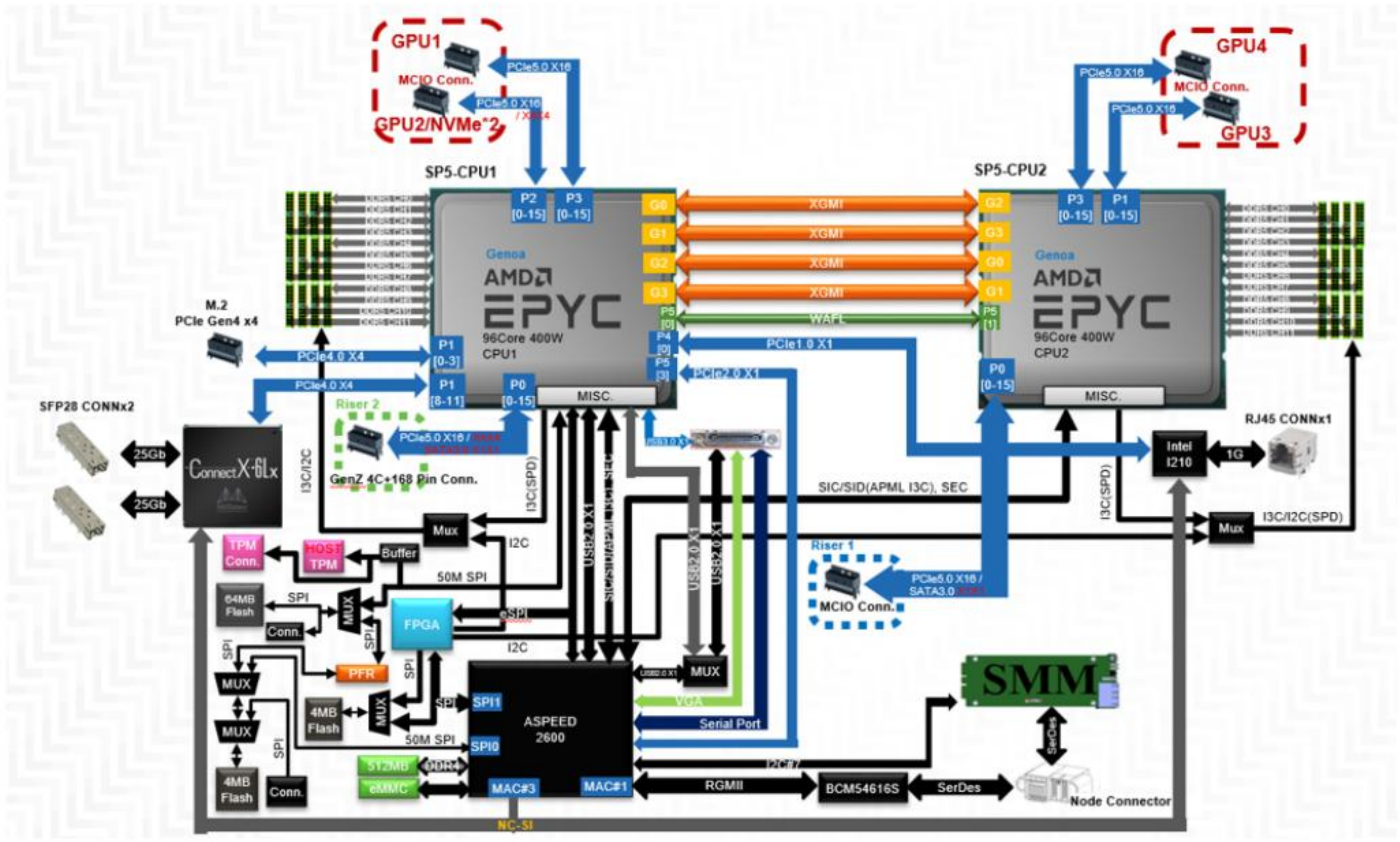
ThinkSystem SD665 V3 system components

System diagram, riser adapter, PCIe adapter, and storage kit

SD665 V3 block diagram



SD665 V3 specifications

Scroll down for more information

Features	Descriptions
Form Factor	1U2N with 6U chassis
Processor	Two AMD EPYC 9004 Series processors (codename: Genoa) per node. Supports processors with up to 96 cores, core speeds of up to 3.5 GHz, and TDP ratings of up to 360 W, cTDP up to 400 W.
Memory	• 48 memory slots with 12 channels per processor • Support for 128 GB 3DS RDIMMs and up to 3 TB per node

Storage	Each node supports one of the following:
	• Up to four 7 mm 2.5-inch drive bays supporting SATA or PCIe 5.0 NVMe drives (but not both)
	  • Four 7 mm drive bays without a PCIe slot
	  • Two 7 mm drive bays with one PCIe slot
	  • No 7 mm drive bays with two PCIe slots (support for an M.2 drive)
	• One 15 mm 2.5-inch drive bay supporting a PCIe 5.0 NVMe drive
	  • Two 15 mm drive bays without a PCIe slots
	  • One 15 mm drive bay with one PCIe slot
	  • No 15 mm drive bays with two PCIe slots (support for an M.2 drive)
	Each node also supports one high-speed M.2 SSD with a PCIe 4.0 x4 connection installed on an M.2 adapter mounted on top of the front processor

Note: For more information about system specifications, refer to Lenovo Press

Lenovo

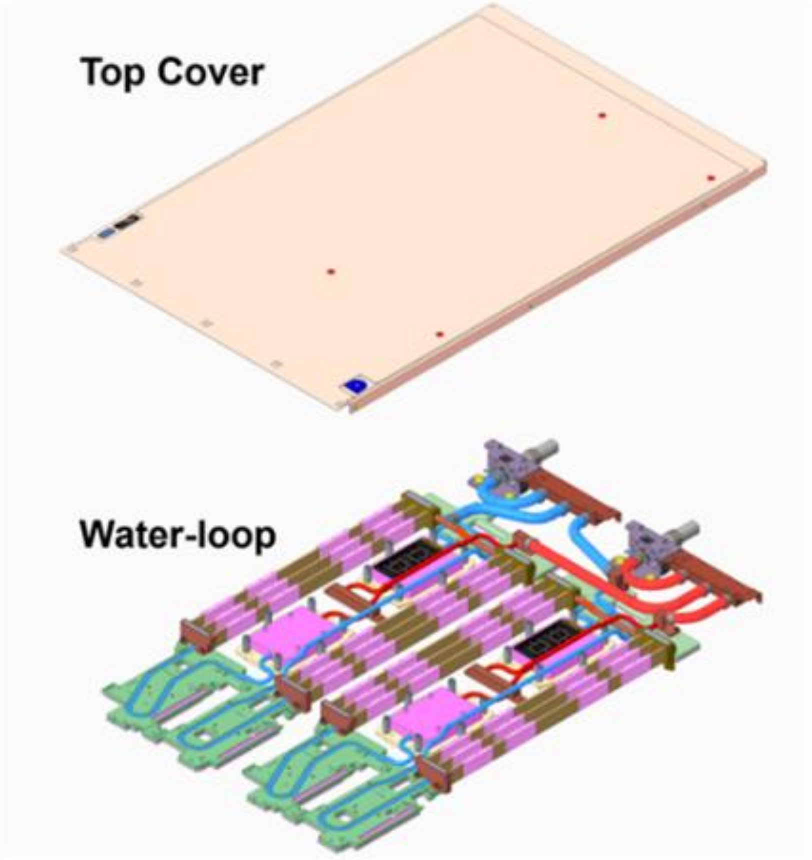
SD665 V3 specifications

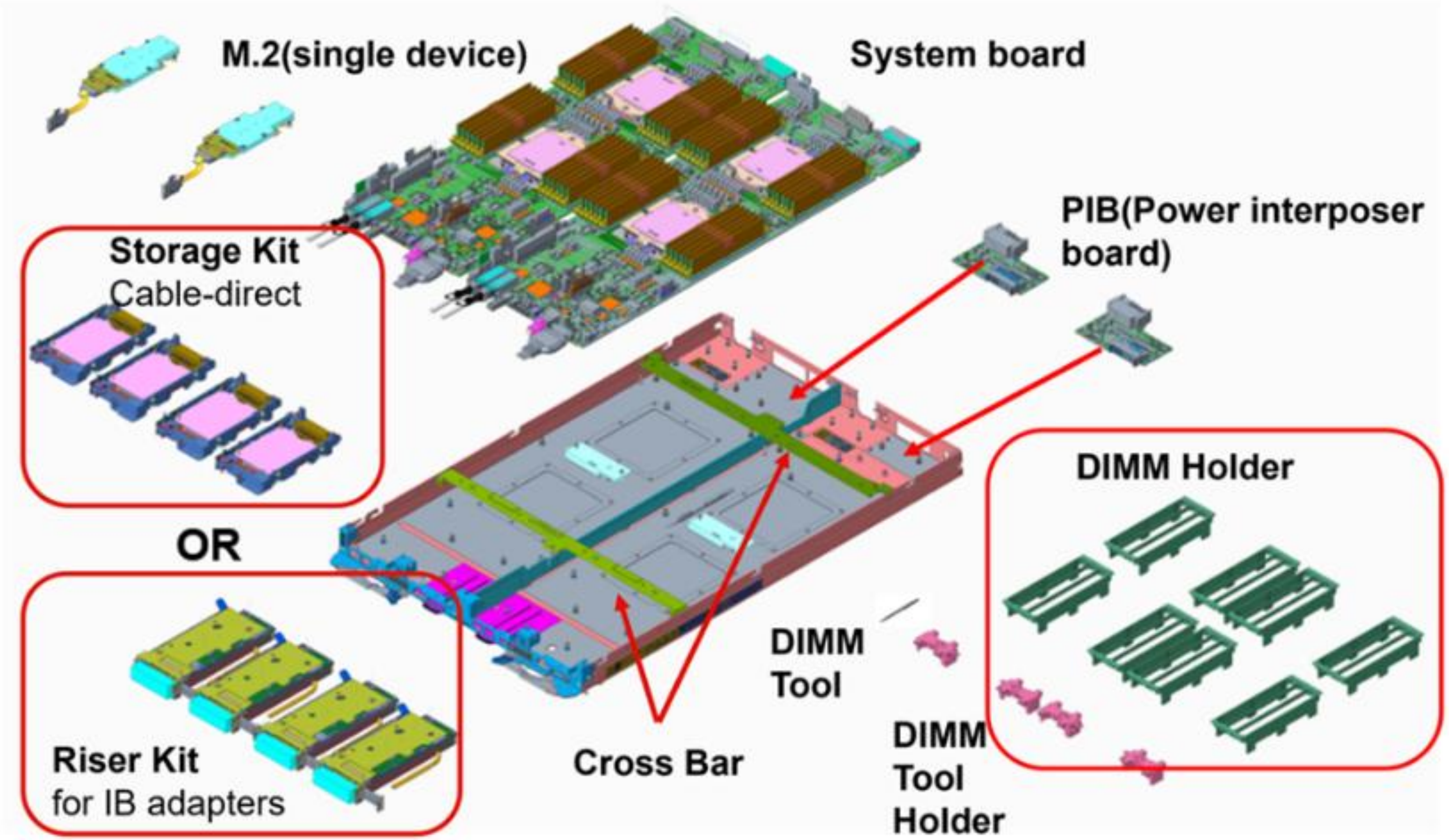
Scroll down for more information

Features	Descriptions
	mounted on top of the front processor Note: Configurations with two PCIe slots only support the M.2 drive for internal storage.
RAID	OS level RAID
Cooling	Direct Water Cooling (DWC)
Networking	• NVIDIA ConnectX6-Lx 25 Gbe LOM (two SFP28 ports) • Intel I210 1 Gb LOM (one RJ45) Notes: Selectable shared NIC support on either SFP port 1 or RJ45
Management	• XCC • Redfish support
Front I/O	• KVM dongle to support one VGA port, one USB 3.1 port, and one serial port • Optional external diagnostic port (Pong connector)
PCIe	• One or two PCIe 5.0 x16 slots with a low-profile form factor per node Notes: PCIe slot support and the drive kit are mutually exclusive

Note: For more information about system specifications, refer to Lenovo Press

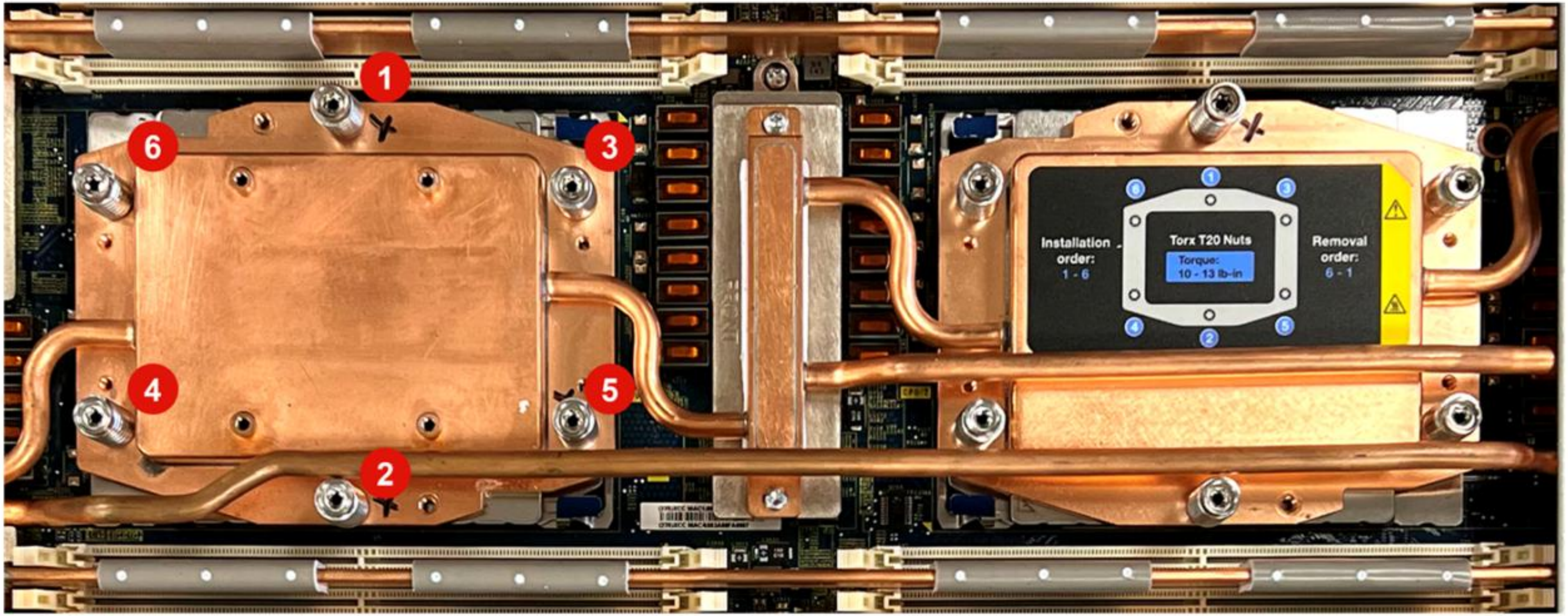
SD665 V3 node tray exploded view





SD665 V3 processor cold plate

A Torx T20 screwdriver is required to remove the processor cold plate. Follow the screw sequence specified on the processor cold plate label. The screw installation sequence is 1→2 → 3→4→5→6.



Torque requirements for SD665 V3 water loop screws

The torque required to fully tighten or remove the screws on major components:

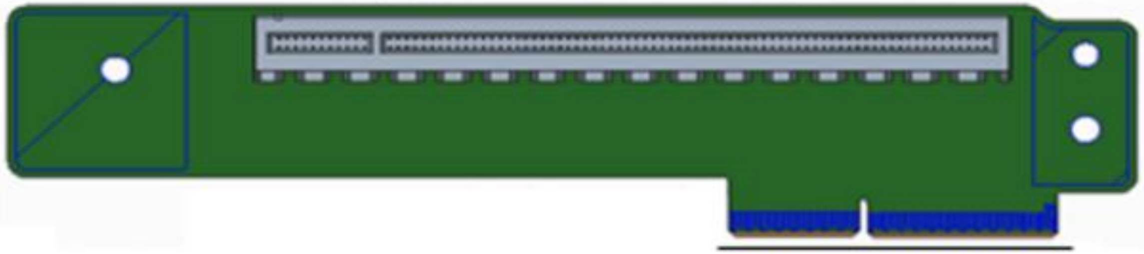
Part	Tool	Torque
Processor cold plate	T20	1.12-1.46 newton-meters, 10-13 inch-pounds
Water loop	T10	0.5-0.6 newton-meters, 4.5-5.5 inch-pounds
System board	M3 screw	0.5-0.6 newton-meters, 4.5-5.5 inch-pounds

SD665 V3 PCIe riser assembly

The SD665 V3 supports the following two PCIe riser configurations:

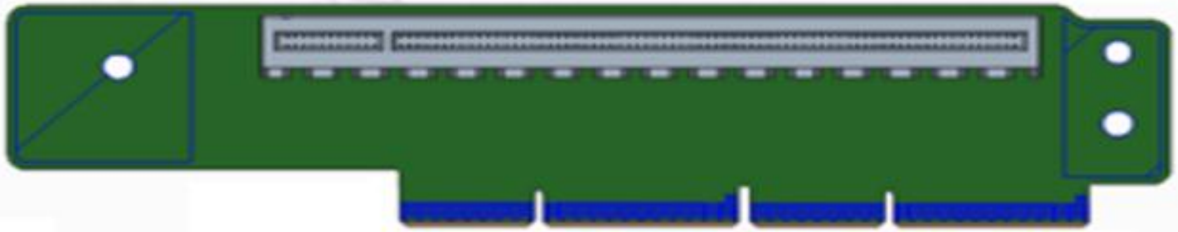
Slot1

Riser 1, x16 Gen5, Gen-Z 2C connector
- PCIe 5.0 support from CPU2 via cabling



Slot2

Riser 2, x16 Gen5, Gen-Z 4C+ connector
- PCIe 5.0 support from CPU1

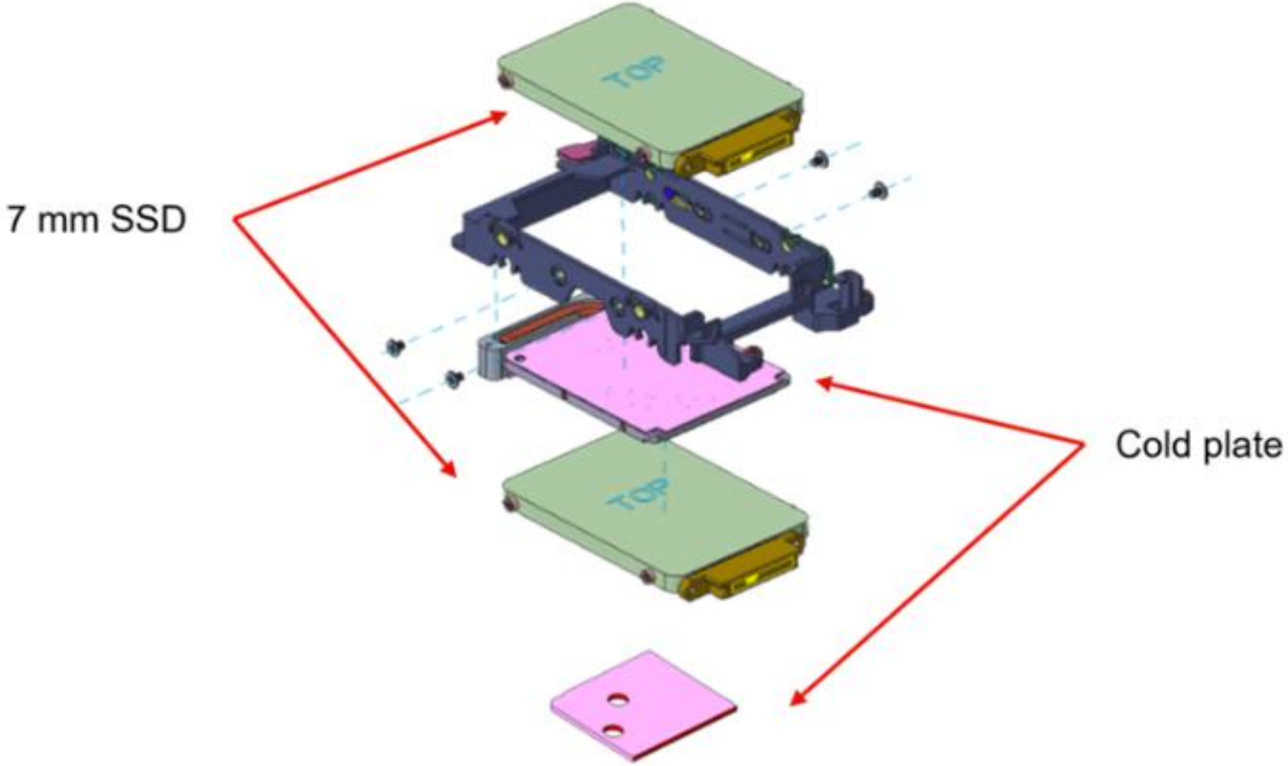


SD665 V3 PCIe adapters

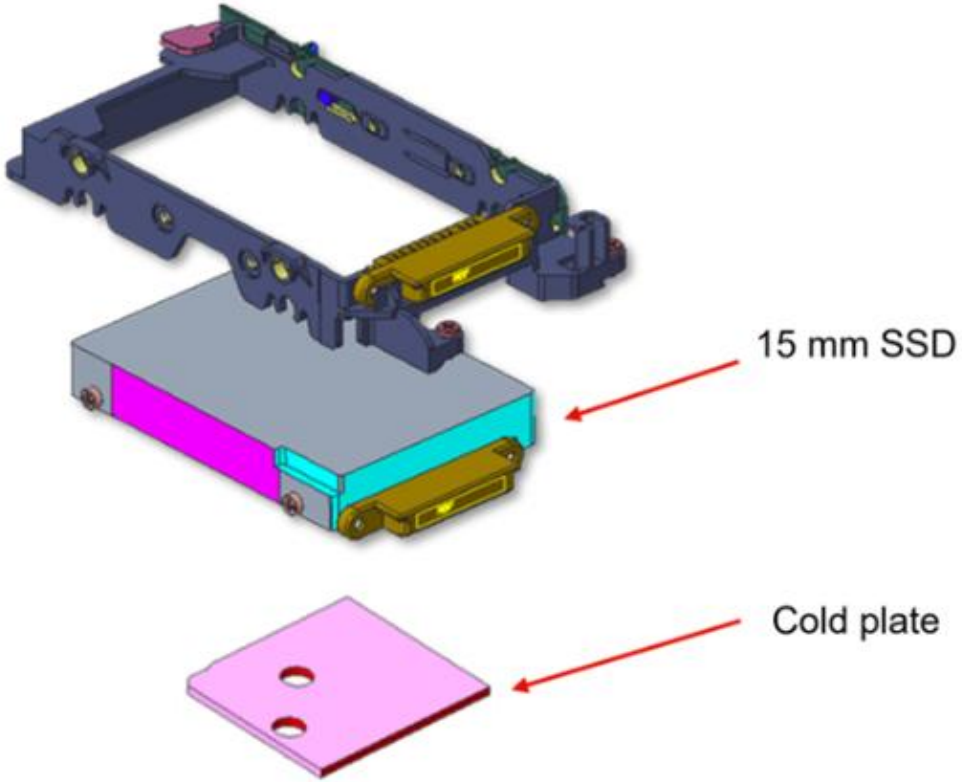
The SD665 V3 supports the following PCIe adapters:

- ThinkSystem NVIDIA ConnectX-7 NDR OSFP400 1-port PCIe 5.0 x16 InfiniBand Adapter DWC
- ThinkSystem NVIDIA ConnectX-7 NDR OSFP400 1-port PCIe 5.0 x16 InfiniBand Adapter (SharedIO) DWC
- ThinkSystem NVIDIA ConnectX-7 NDR200/HDR QSFP112 2-port PCIe 5.0 x16 InfiniBand Adapter (SharedIO) DWC
- ThinkSystem Mellanox ConnectX-6 HDR/200GbE QSFP56 1-Port PCIe 4.0 VPI Adapter (SharedIO) DWC

SD665 V3 SSD storage drive assembly

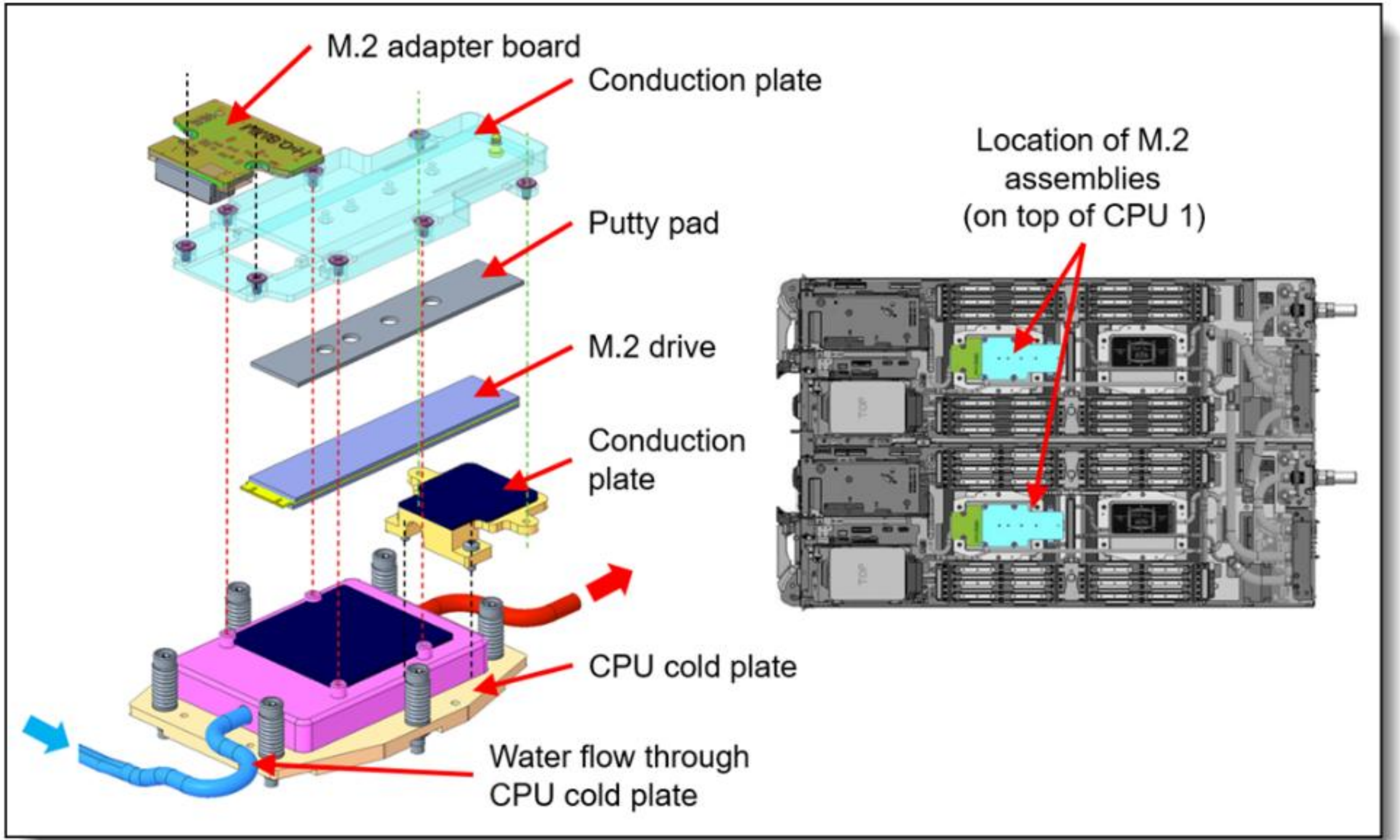


Two 7 mm NVMe or SATA SSDs
with a cold plate



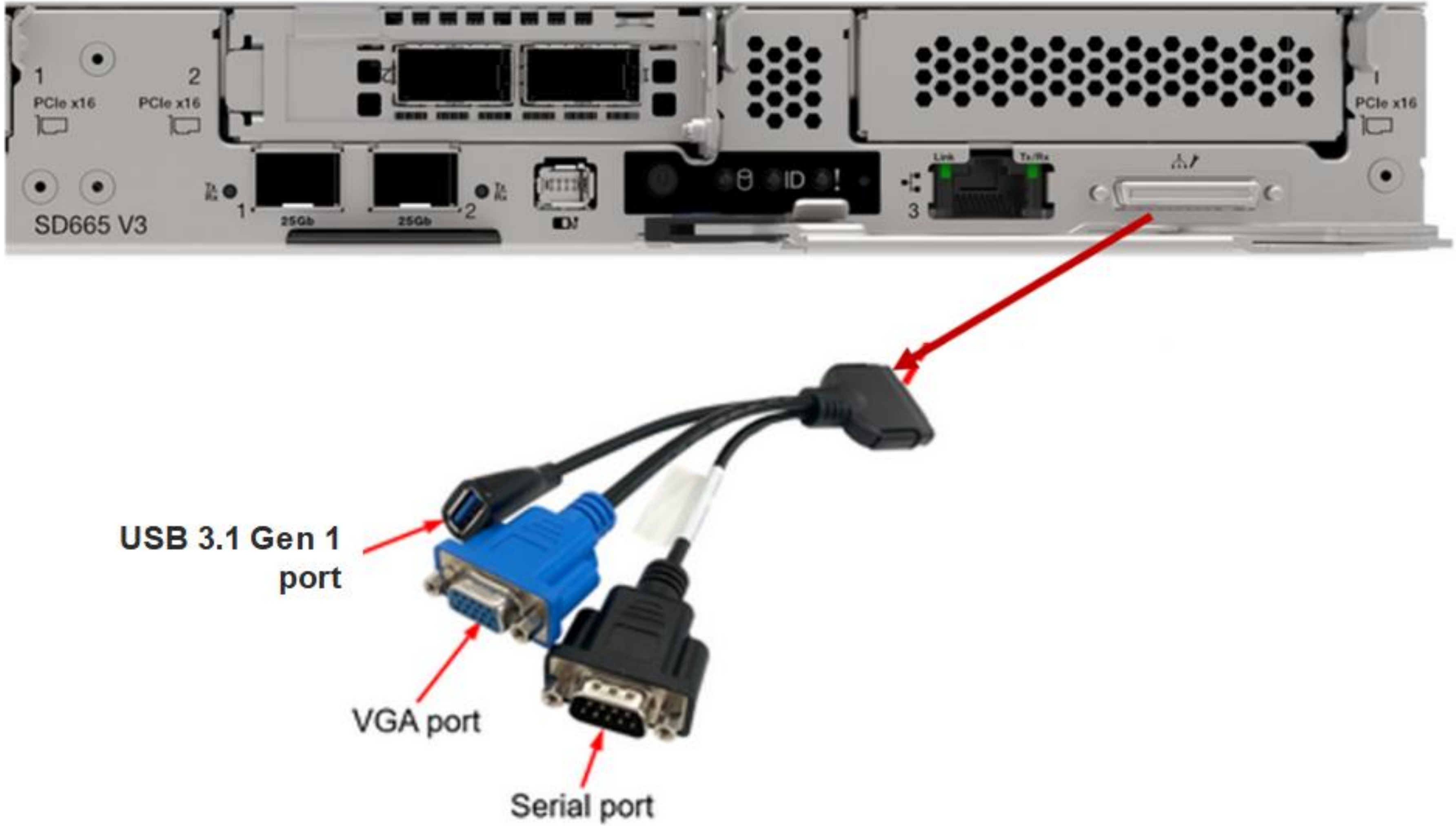
One 15 mm NVMe SATA
SSD with a cold plate

SD665 V3 NVMe M.2 storage assembly



KVM breakout cable

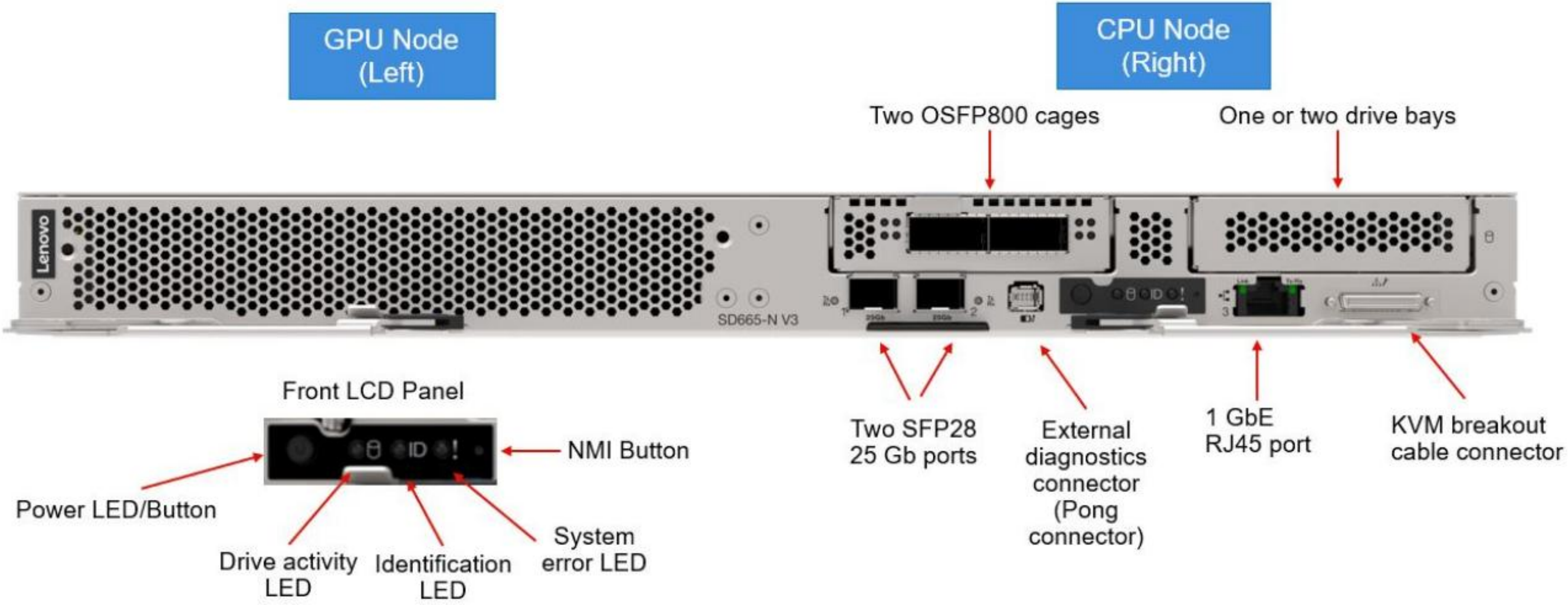
The SD665 V3 supports a local console through a KVM breakout cable. The cable connects to the port on the front of the server, as shown in the following figure.



Product overview

The ThinkSystem SD665-N V3 (MT 7DAZ) is based on the Neptune DWC platform. The SD665-N V3 has one CPU compute node and one GPU node mounted on a shared 1U tray with a water loop covering all the major heat sources from both nodes.
The CPU node used in the SD665-N V3 is the same as the one used in the SD665 V3. The GPU node supports four NVIDIA H100 Tensor Core GPUs.

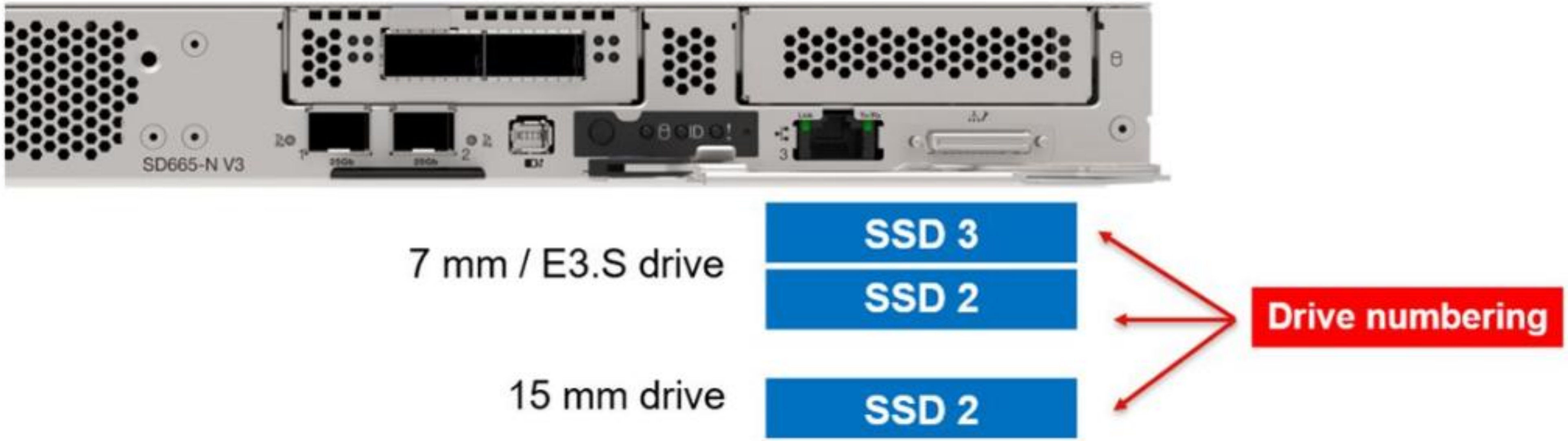
Front view


GPU Node (Left)
CPU Node (Right)
Two OSFP800 cages
One or two drive bays
Lenovo
SD665-N V3
Front LCD Panel
NMI Button
Power LED/Button
Drive activity LED
Identification LED
System error LED
Two SFP28 25 Gb ports
External diagnostics connector (Pong connector)
1 GbE RJ45 port
KVM breakout cable connector

Storage configurations

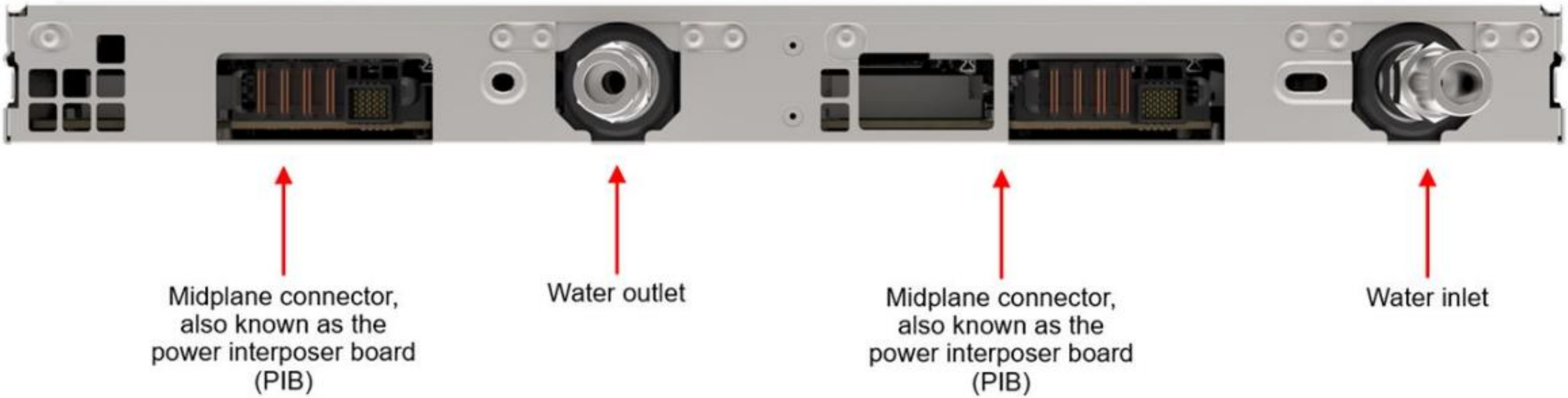
The SD665-N V3 supports the following front and internal storage configurations:

- Two front 7 mm NVMe SSD drives or one 15 mm U.2/U.3 NVME drive
- Two front E3.S EDSFF drives
- An internal NVMe M.2 assembly supporting a single M.2 drive

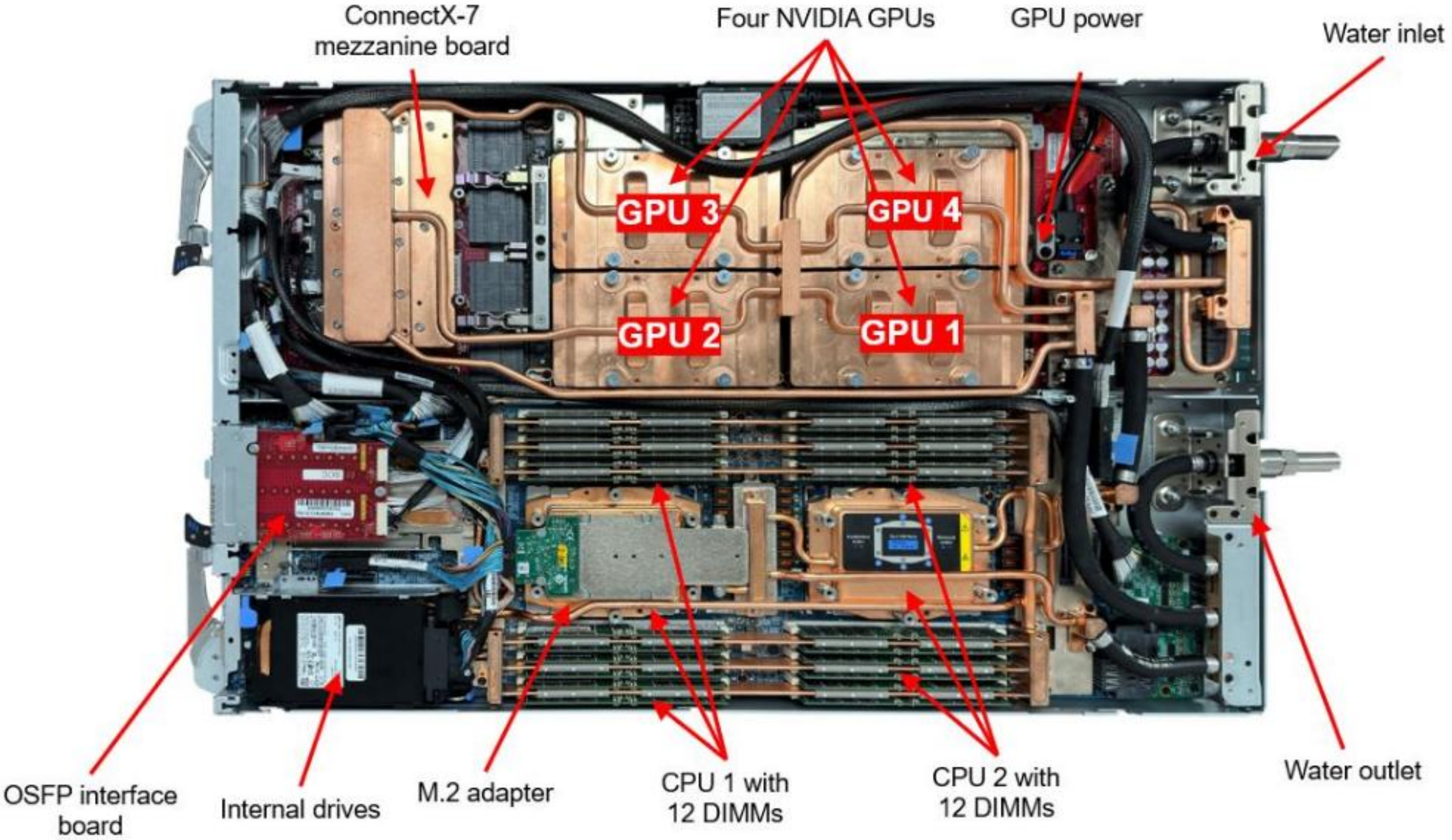


> **Note**: The SD665-N V3 drives are not hot-swappable. Before replacing a drive, power off the node, remove the trays from the enclosure, and remove the top cover.

Rear view


Midplane connector, also known as the power interposer board (PIB)
Water outlet
Midplane connector, also known as the power interposer board (PIB)
Water inlet

Inside view


ConnectX-7
mezzanine board
Four NVIDIA GPUs
GPU power
Water inlet
GPU 3
GPU 4
GPU 2
GPU 1
OSFP interface
board
Internal drives
M.2 adapter
CPU 1 with
12 DIMMs
CPU 2 with
12 DIMMs
Water outlet

Processor configurations

The SD665-N V3 CPU node supports one or two 4th generation AMD EPYC processors. A field upgrade from one processor to two processors is currently not supported.

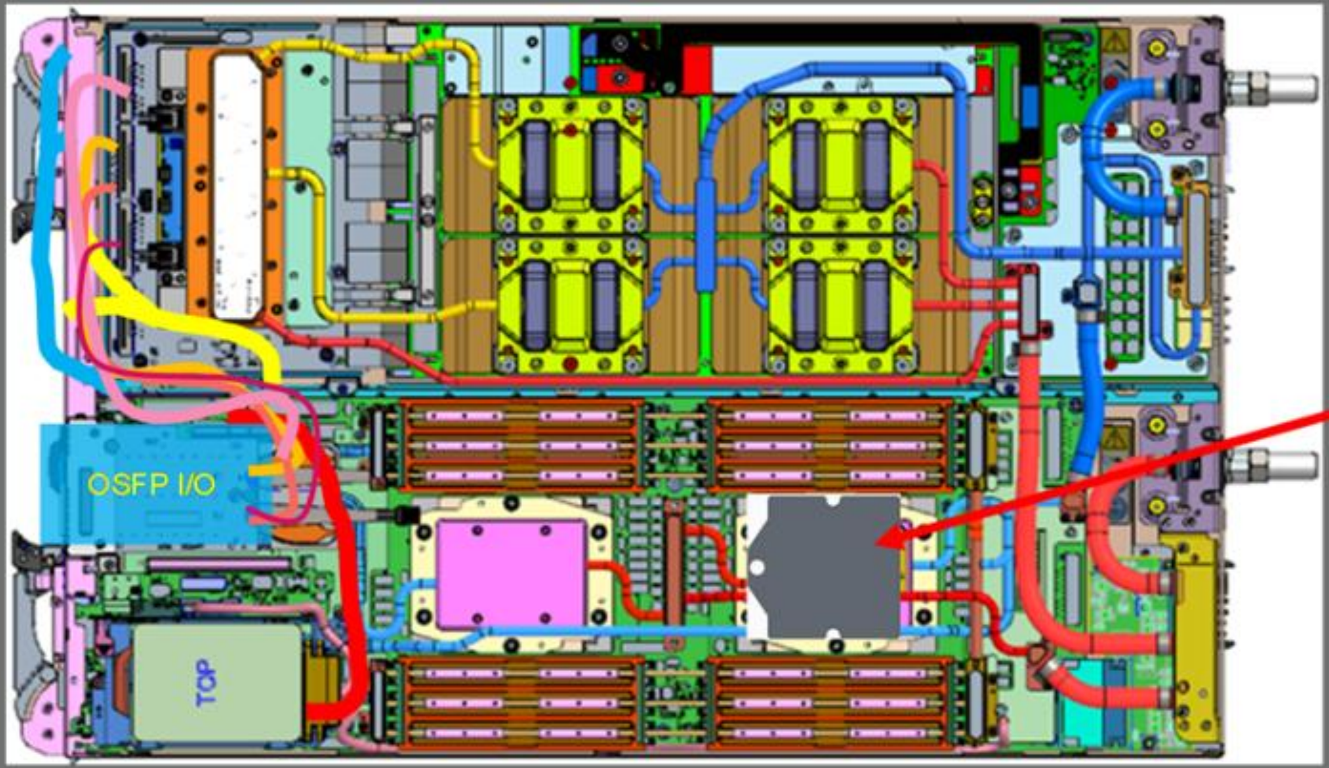
Different MCIO PCIe cables are used with the two configurations:

- For one-processor systems, use two MCIO PCIe x8x8x8x8 Y-cables
- For two-processor systems, use four MCIO PCIe x16 cables

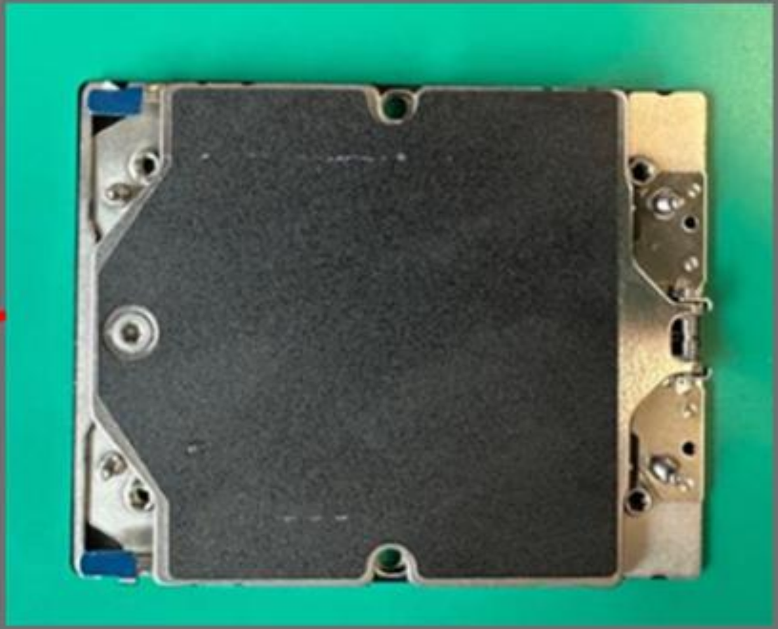
If the node is set up in a one-processor configuration, the second processor slot must be covered by a mylar shield. (Click HERE to see an image.)

Four MCIO PCIe x16 cables

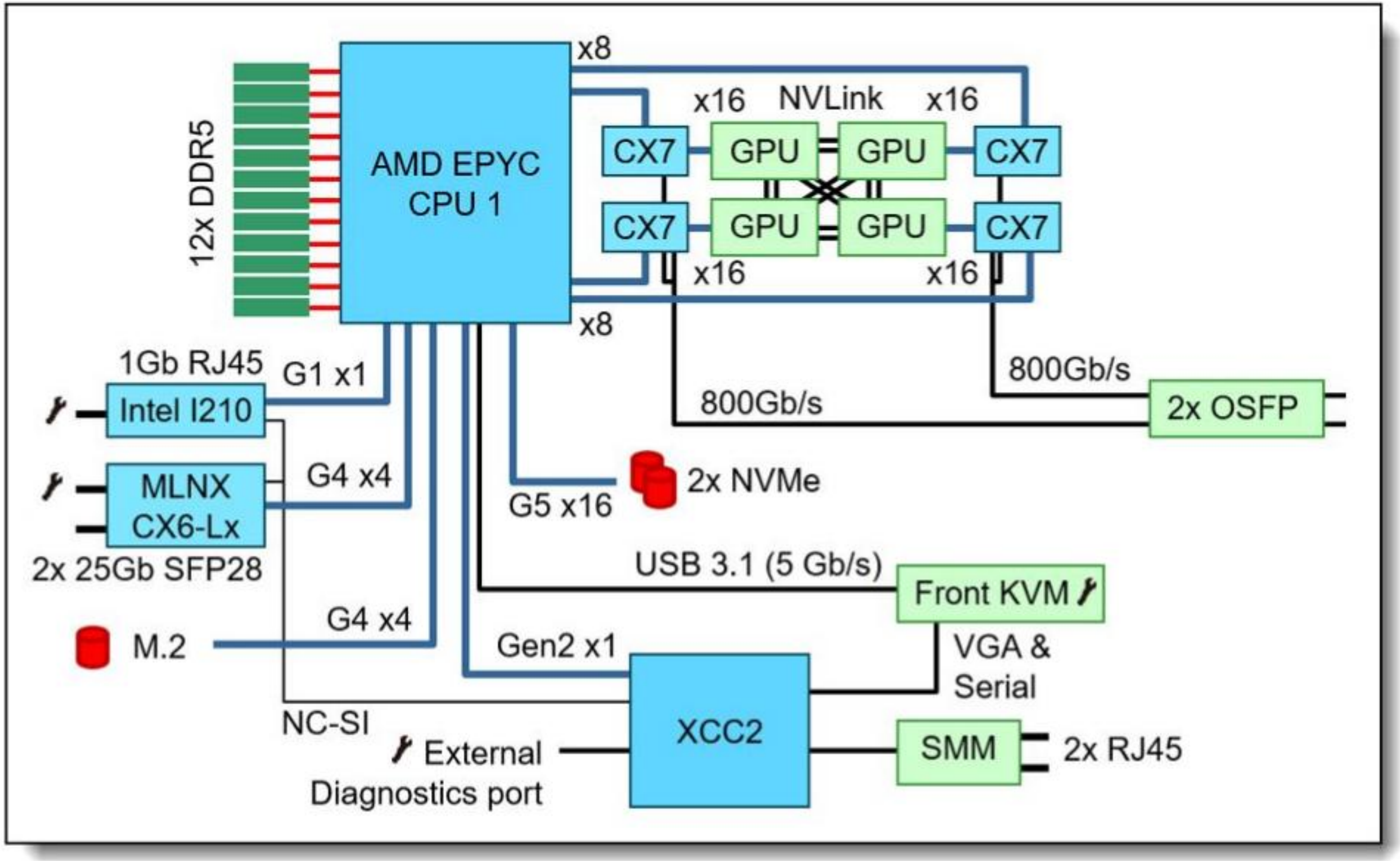
Two MCIO PCIe x8x8x8x8 Y-cables



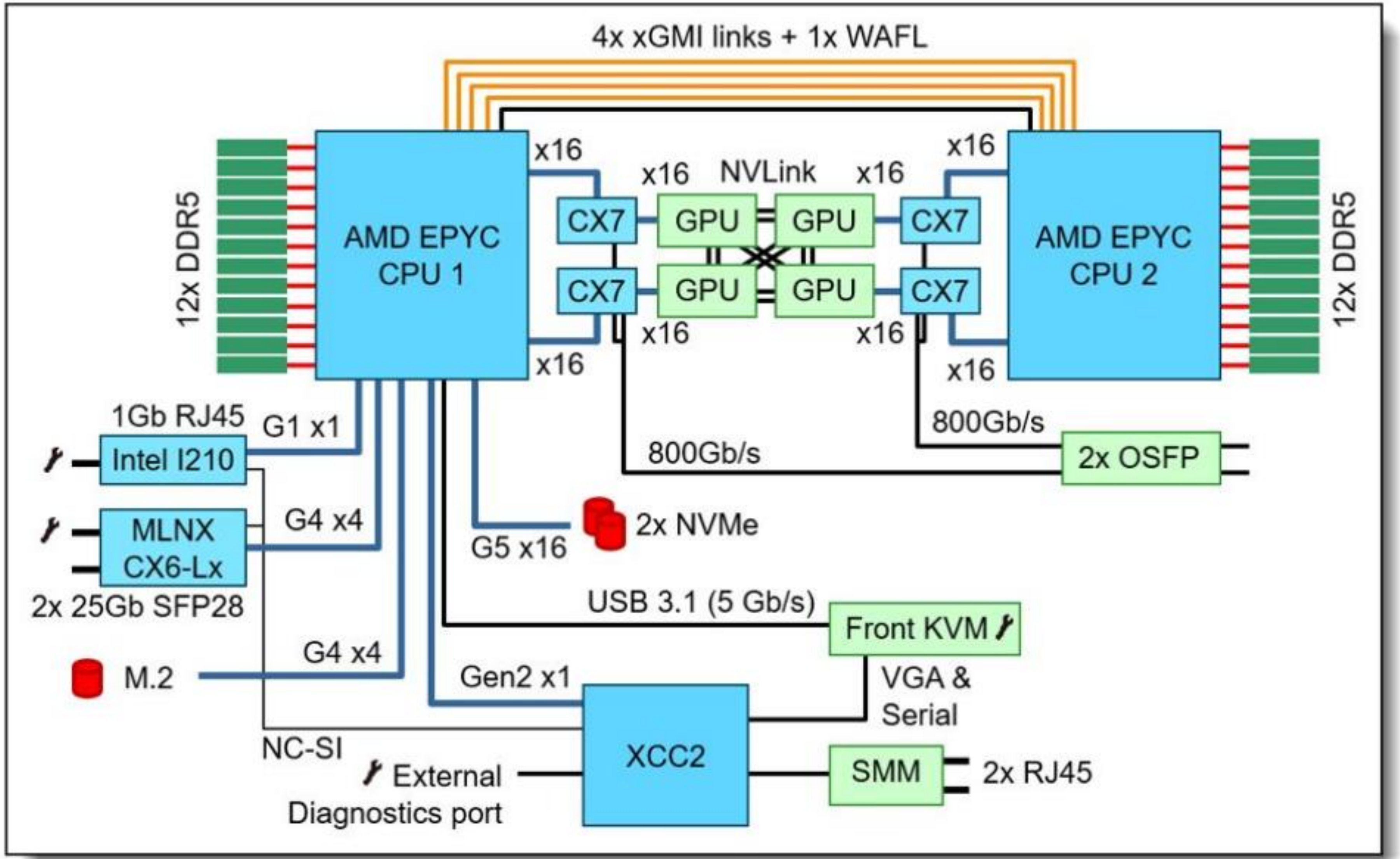
2nd CPU slot



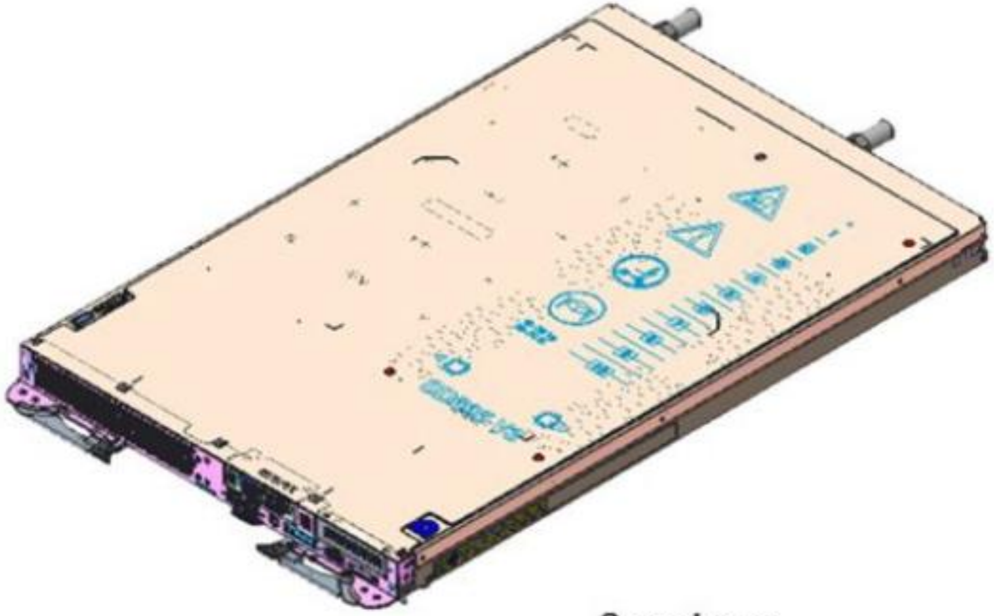
SD665-N V3 block diagram – one processor



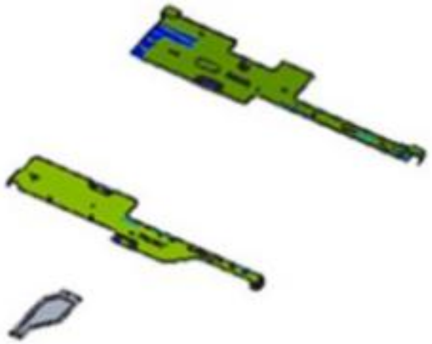
SD665-N V3 block diagram – two processors

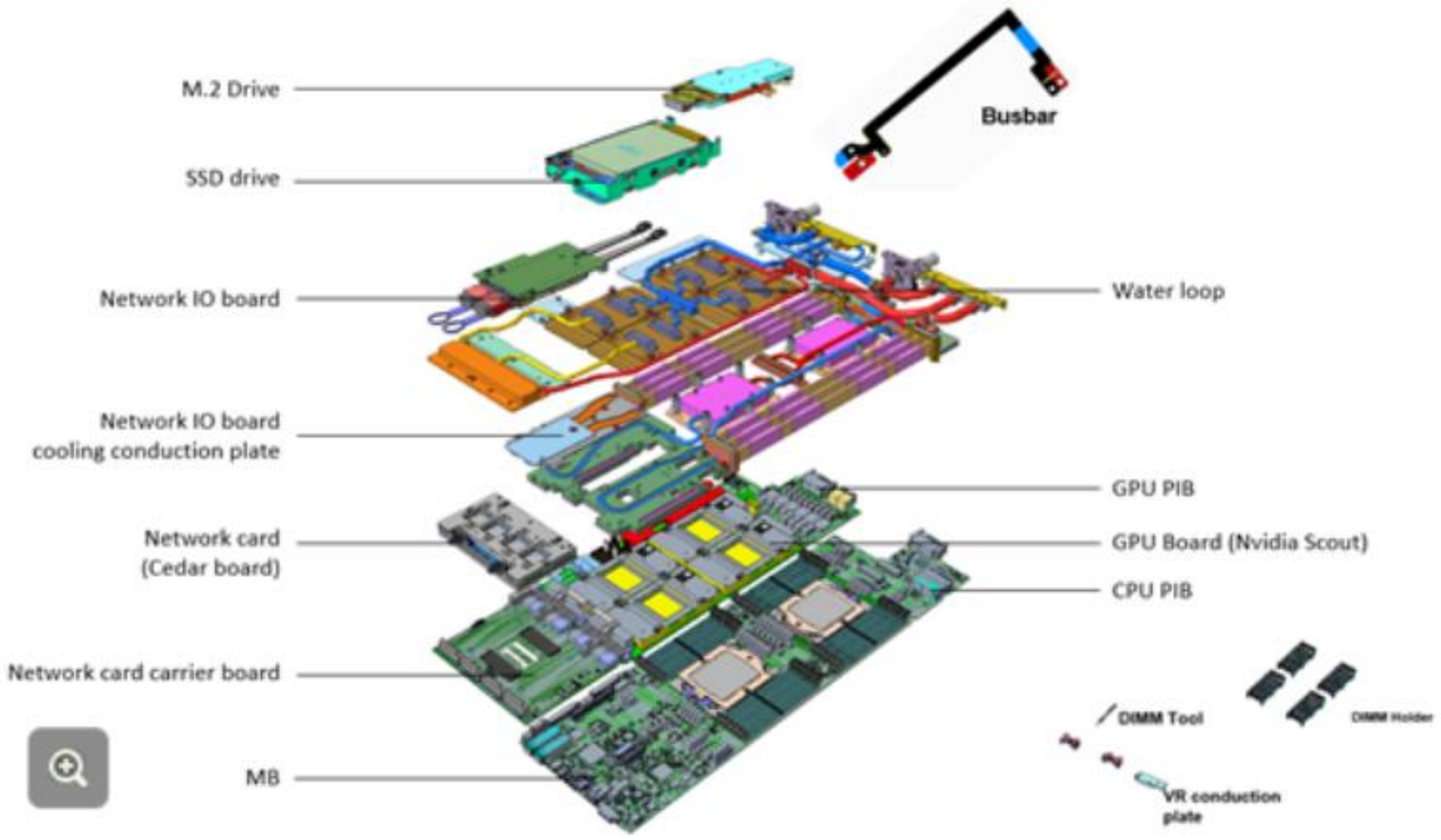


Exploded view



Cross brace





OSFP NDR I/O interface board

The SD665-N V3 includes an I/O mezzanine board containing four NVIDIA ConnectX-7 VPI network controllers.
The mezzanine board has two connectors, allowing an OSFP board to be attached via cables as shown in the following figure. The server makes use of OSFP-DD (double-density) connections to double the bandwidth from 400 Gb/s to 800 Gb/s per physical port.